# Loop-Mediated Isothermal Amplification for Detection of Nucleic Acids



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## ABSTRACT

Sequence-specific isothermal nucleic acid amplification techniques are ideally suited for use in molecular diagnostic applications because they do not require thermal cycling equipment and the reactions are typically fast. One of the most widely cited isothermal techniques is termed loop-mediated isothermal amplification (LAMP). This protocol allows amplification times as fast as 5 to 10 min. Furthermore, various methodologies to detect amplification have been applied to LAMP to increase its utility for the point-of-care market. Basic LAMP protocols are provided herein for detection of specific DNA and RNA targets, along with a method to perform multiplex LAMP reactions, permitting even greater flexibility from this powerful technique. *Curr. Protoc. Mol. Biol.* 105:15.14.1-15.14.14. © 2014 by John Wiley & Sons, Inc.

Keywords: LAMP • isothermal amplification • molecular diagnostics

## INTRODUCTION

Recent years have seen a dramatic rise in the abundance and availability of nucleic acid sequence information. This expansion has enabled the use of DNA and RNA amplification techniques for specific detection, harnessing the complexity inherent in genetic material for the purpose of targeted identification. Molecular diagnostic techniques utilizing nucleic acids were pioneered through use of the polymerase chain reaction (PCR), which remains the predominant method in the field due to its robustness, sensitivity, and familiarity. However, the growing use of these molecular diagnostic methods has emphasized speed and simplicity as key criteria for adoption in point-of-care and field applications, and isothermal amplification techniques are well-suited for these uses (isothermal methods and point-of-care applications are reviewed in Niemz et al., 2011; Njiru, 2012). By their nature, isothermal amplification methods require only a single temperature, avoiding the need for expensive thermal cycling equipment and potentially even electrical power, depending on incubation temperature and heating. Additionally, by constant incubation and amplification, no temporal constraints from defined cycles are implied, resulting in amplification reactions as rapid as 5 min. Perhaps the most widespread isothermal method is loop-mediated isothermal amplification (LAMP). Since its publication in 2000 (Notomi et al., 2000), LAMP has been applied to diagnostic detection of hundreds of pathogens in clinical, plant, food, and animal samples. LAMP presents a simple, robust, and flexible platform for molecular diagnostics.

This unit describes the basic methods and materials for performing LAMP reactions using DNA (Basic Protocol) and RNA (Alternate Protocol 1) targets. Alternate Protocol 2 provides a guide for multiplexed LAMP reactions using labeled primers. While simple, LAMP can require experience and practice to optimize, so helpful strategies are provided for implementation and troubleshooting of LAMP reactions.



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*BASIC PROTOCOL*

## LAMP REACTIONS WITH DNA TEMPLATES

This protocol describes a basic LAMP reaction using DNA template and six LAMP primers (four core, two loop) (Notomi et al., 2000; Nagamine et al., 2002). The LAMP principle is shown in Figure 15.14.1. Reactions can be monitored in real time by fluorescence, turbidity, luminescence, or metal-sensitive dye (Mori et al., 2004; Goto et al., 2009; Gandelman et al., 2010). Alternatively, reactions can be analyzed after amplification by visual inspection of fluorescence, color, or turbidity, or using gel electrophoresis, lateral flow devices, or nanoparticle detection (Tomita et al., 2008; Murray et al., 2013; Seetang-Nun et al., 2013).

Primers for LAMP reactions can be chosen for any target nucleic acid, with either high specificity (detecting only one target) or universality (detecting many related targets). Primer design may be performed manually, but software is available for efficient primer design using various recommended settings (e.g., length of amplicons and primers, avoiding AT richness for ends of FIP/BIP, proper $T_m$; Tomita et al., 2008). LAMP primers for *BRCA1* were designed using PrimerExplorer v4 (Eiken Chemical), but LAMP Designer (Premier Biosoft) is another available software option. LAMP is prone to non-template amplification due to the amount and number of primers, so care should be taken to avoid primer dimers or other effects. Software solutions will automatically avoid obvious problem sequences and are highly recommended.

If possible, order multiple complete primer sets for validation and optimization of the reaction, including controls for both positive and negative performance. HPLC or gel purification of primers is generally not required, but can improve specificity and reproducibility for diagnostic and multiplex applications (Tomita et al., 2008).

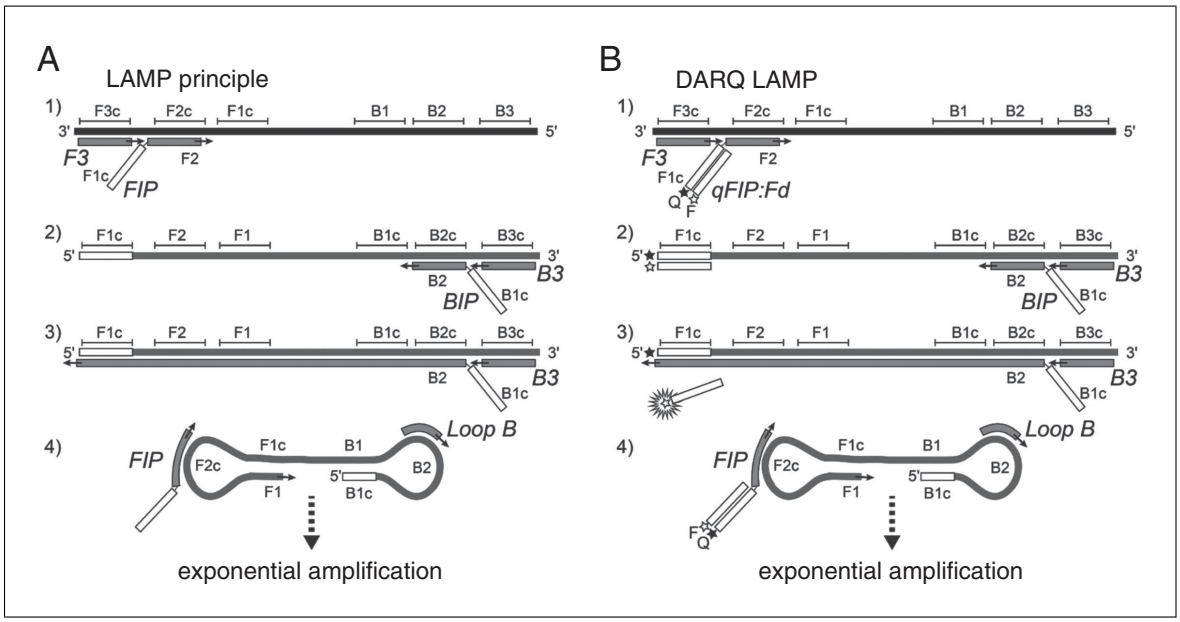


**Figure 15.14.1** LAMP amplification scheme. (**A**) Standard LAMP amplification. (1) Initiation at F end of target sequence via FIP (primers in italics) and displacement of nascent strand by synthesis initiating at F3. (2,3) Synthesis and displacement at B end of target resulting in (4), the seed structure for exponential LAMP amplification. (**B**) DARQ LAMP amplification using qFIP:Fd duplex primer with a fluorophore (F) and dark quencher (Q). LAMP proceeds as in (A), but with displacement of Fd in (3) producing the fluorescent signal. Additional qFIP:Fd duplex participates in downstream amplification and is subsequently displaced to produce additional fluorescence during exponential amplification (4). More detailed illustration of LAMP products can be found in Notomi et al. (2000) and Nagamine et al. (2002).

***Materials***

Oligonucleotide primers (typically, 100 μM stocks in $H_2O$ or 1× TE buffer)

*For human breast cancer 1 (BRCA1; 5′ to 3′):*
F3: TCCTTGAACTTTGGTCTCC
B3: CAGTTCATAAAGGAATTGATAGC
FIP: ATCCCCAGTCTGTGAAATTGGGCAAAATGCTGGGATTATA-GATGT
BIP: GCAGCAGAAAGATTATTAACTTGGGCAGTTGGTAAGTAAATG-GAAGA
Loop F: AGAACCAGAGGCCAGGCGAG
Loop B: AGGCAGATAGGCTTAGACTCAA

Test sample
Control template DNA (e.g., HeLa genomic DNA, New England Biolabs, 0.1 mg/ml)
LoopAmp Extraction kit (Eiken Chemical) or other method for DNA extraction (optional)
2× LAMP reaction buffer (see recipe)
Dye for visual detection (optional), such as:

50 μM SYTO-9 (see recipe, 2 μM final) or other intercalating dsDNA dye
625 μM calcein (see recipe, 25 μM final)
3 mM hydroxynaphthol blue (see recipe, 120 μM final)
SYBR Green (Life Technologies/BioRad, 0.1× to 1× final)

Strand-displacing DNA polymerase (e.g., *Bst* 2.0, large fragment, New England Biolabs, 8,000 U/ml)
Molecular-biology grade water
5 M betaine solution (see recipe; optional)

Appropriate plates (with sealing film) or tubes for instrumentation
65°C heating block, thermal cycler, or water bath
Fluorescence or turbidity instrument for real-time detection (e.g., LA-320c, SA-Scientific; ESE-Quant, Qiagen; CFX, Bio-Rad)

***Prepare LAMP primer mixes***

1. From primer stock solutions, prepare two 25× mixes of core primers and loop primers, or a single 20× or 10× mix of all six primers.

   For 100 μl of 25× core primers:

   40 μl 100 μM FIP
   40 μl 100 μM BIP
   5 μl 100 μM F3
   5 μl 100 μM B3
   10 μl $H_2O$.

   For 100 μl of 25× loop primers:

   10 μl 100 μM Loop F
   10 μl 100 μM Loop B
   80 μl $H_2O$.

   Alternatively, for 100 μl of 20× LAMP primers:

   32 μl 100 μM FIP
   32 μl 100 μM BIP
   4 μl 100 μM F3

4 μl 100 μM B3
8 μl 100 μM Loop F
8 μl 100 μM Loop B
12 μl $H_2O$.

*Suggested final concentrations are 1.6 μM FIP, 1.6 μM BIP, 0.2 μM F3, 0.2 μM B3, 0.4 μM Loop F, and 0.4 μM Loop B. Individual primer stocks as well as mixes should be stored at −20°C for long-term stability (up to 3 years or longer).*

***Prepare samples***

2. If desired, perform nucleic acid extraction on test samples to avoid inhibition of the reaction (e.g., LoopAmp Extraction kit, Eiken Chemical).

   *Compared to PCR, LAMP is more tolerant of many culture media components and samples (e.g., MEM, vitreous humor), but high levels of material from clinical (e.g., blood, sputum, fecal) or plant (e.g., humic acid) samples can affect the reaction (Kaneko et al., 2007). Typical nucleic acid extraction techniques (e.g., Qiagen columns, DNAzol, phenol/chloroform extraction, ethanol precipitation) are recommended if samples are highly impure or may interfere with detection methods (e.g., whole blood).*

3. For improved specificity and amplification efficiency with some templates, denature the template or sample prior to amplification by heating to 95°C for 5 min and then placing on ice.

   *Denaturation must be performed prior to addition of Bst polymerase and initiation of amplification, as the polymerase is not sufficiently heat-stable.*

***Perform LAMP reactions***

4. Prepare sample and control reaction mixtures on ice, as follows:

   12.5 μl 2× LAMP reaction buffer
   1 μl 25× primer mix
   1 μl 25× loop mix
   1 μl dye for visual detection (optional)
   1 μl 8 U/μl *Bst* 2.0 DNA polymerase
   1 μl test sample or 10 ng control template DNA
   7.5 μl $H_2O$.

   For consistency, perform each reaction in duplicate or triplicate 25-μl reactions. Include negative control reactions using water in place of sample or template DNA.

   *Keep reactions on ice, as nonspecific room-temperature activity of Bst DNA polymerase will adversely affect reactions (Tanner et al., 2012). For room-temperature, high-throughput, or field setup, Bst 2.0 WarmStart (New England Biolabs) can be used (Poole et al., 2012).*

   *Addition of intercalating dye causes some inhibition of the amplification reaction, but SYTO-9 has been demonstrated to produce minimal inhibition (Monis et al., 2005; Njiru et al., 2008).*

   *If desired, 0.4 to 1.2 M betaine (Sigma) can be added to reduce nonspecific amplification for certain primer sets. Efficacy of this addition can be examined by running positive and negative reactions with and without betaine.*

5. Mix components by pipetting or vortexing, then briefly pulse-spin in a microcentrifuge.

6. For quantitation, run a series of template or sample concentrations.

   *LAMP can typically provide a dynamic range of $10^4$ to $10^8$, with maximum sensitivity of 1 to 1000 copies, depending on the efficiency of amplification and level of non-template (negative) amplification. Measure both parameters with control reactions before or during the test sample run.*

7. Place the reaction directly into a preheated 65°C chamber and incubate for the desired time.

   *Threshold times using fluorescence detection are typically 3 to 30 min depending on the amount of template and efficiency of amplification. If using real-time turbidimetry, threshold times will be slower, e.g., 10 to 60 min (Mori et al., 2004).*

8. Perform endpoint detection after 45 to 60 min incubation using agarose gel electrophoresis, alternative methods such as nanoparticles (Seetang-Nun et al., 2013), or visual detection using a UV or light box.

   CAUTION: *When opening reaction vessels for analysis, extreme care should be taken to avoid contamination of laboratory areas and equipment. Tubes or plates should only be opened in areas separate from those where reactions are prepared, ideally in a laminar flow hood using separate pipets and equipment. Due to the high level of amplification, LAMP is extremely sensitive to even small amounts of carryover contamination (see Critical Parameters and Troubleshooting).*

   *Conditions and parameters for optimizing LAMP reactions, including buffer and additives, temperature, and reagent concentrations, are discussed in Critical Parameters and Troubleshooting.*

## RT-LAMP REACTIONS WITH RNA TEMPLATES

***ALTERNATE PROTOCOL 1***

A benefit to LAMP reactions is its inherent compatibility with reverse transcription (RT). First, unlike PCR reactions, LAMP is performed in buffers containing high levels of magnesium (typically 8 mM), optimal for RT activity. Second, LAMP amplicons are diagnostic in function and thus quite short (<200 bp), obviating the need for long incubations to synthesize long, contiguous cDNAs. These factors and the lack of a heat denaturation step allow for true one-step RT-LAMP reactions, wherein RNA template can be added in the presence of reverse transcriptase, strand-displacing DNA-dependent DNA polymerase (*Bst*), and LAMP primers and incubated at reaction temperature (Notomi et al., 2000). If desired, a separate cDNA synthesis step can be performed, although this is typically not advantageous.

*Bst* and other DNA polymerases have been demonstrated to have RT activity, although it is variable and much lower than that of a native RT. However, newer *Bst*-like enzymes used for LAMP (e.g., *Bst* 2.0, New England Biolabs; GspSSD, OptiGene) have significantly increased reverse transcriptase activity and can be used without an additional RT enzyme if desired for compliance, cost, or compatibility concerns. Use of two enzymes (e.g., *Bst* 2.0 and avian myeloblastosis virus [AMV] RT) will provide fastest amplification, but for many amplicons *Bst* 2.0 alone will be sufficiently rapid. With the *HMBS2* primers below, *Bst* 2.0 and AMV used with 10 ng Jurkat RNA gives a LAMP threshold of ~5 min, whereas *Bst* 2.0 alone gives a threshold of ~10 min. A two-enzyme mixture will consistently produce rapid amplification, whereas *Bst* 2.0 alone may exhibit poor performance on some amplicons, but this can be determined empirically for each amplicon, and savings in cost or regulation may outweigh any benefit from using two enzymes. Though similar in preparation, both approaches are given below.

Primers for RT-LAMP are designed with the same considerations as for DNA templates, and the available software options are again recommended (see Basic Protocol). Target cDNA sequence can be input and used for primer design, and, if possible, several distinct primer sets should be validated for optimum performance.

***Additional Materials*** *(also see Basic Protocol)*

Oligonucleotide primers (typically, 100 μM stocks in $H_2O$ or 1× TE buffer)

*For human hydroxymethylbilane synthase (HMBS2; 5′ to 3′)*:
F3: AGAGTGATTCGCGTGGGTA
B3: ACTTCATTCTTCTCCAGGGC
FIP: TCAAACTGCAGGCCAGGGTACCAGCTTGCTCGCATACAGAC
BIP: TTGCTATGTCCACCACAGGGGAAGCTCCTTGGTAAACAGGC
Loop F: GTTGCCACCACACTGTCC
Loop B: CAAGATTCTTGATACTGCACTCTCT

Control template RNA (e.g., Jurkat RNA, Ambion)
Reverse transcriptase (e.g., AMV RT, New England Biolabs 10,000 U/ml)
RNase-free water

1. Prepare primer mixes and sample using the same considerations as for DNA templates (see Basic Protocol, steps 1 and 2). Perform nucleic acid extraction prior to amplification if samples contain inhibitory components.

   *Store reaction components at −20°C for optimum stability and thaw fresh before use in amplification reactions.*

2. Prepare RT-LAMP reactions on ice using similar concentrations of primers and buffers (see Basic Protocol, step 4), with the addition of reverse transcriptase for two-enzyme reactions.

   For 25-μl single-enzyme reaction:

   12.5 μl 2× LAMP reaction buffer
   1 μl 25× primer mix
   1 μl 25× loop mix
   1 μl dye for visual detection (optional)
   1 μl 8 U/μl *Bst* 2.0 DNA polymerase
   1 μl test sample or 10 ng control template RNA
   7.5 μl RNase-free $H_2O$.

   For 25-μl two-enzyme reaction:

   Add 0.5 μl 10 U/μl AMV RT to above mixture
   Decrease $H_2O$ to 7 μl.

   *See Basic Protocol (step 4) for additional considerations, including replicates and controls.*

3. Mix reaction by pipetting or vortexing, then briefly pulse-spin in a microcentrifuge.

4. For quantitation, perform a series of template or sample concentrations.

   *RT-LAMP can typically provide a dynamic range of $10^4$ to $10^8$, with sensitivity to 1 to 1000 copies, depending on the efficiency of amplification and level of non-template (negative) amplification. Measure both parameters with control reactions before or during the test sample run.*

5. Place reaction at 65°C and incubate for the desired time (see Basic Protocol, step 7).

6. Perform endpoint detection after 45 to 60 min (see Basic Protocol, step 8, and Critical Parameters and Troubleshooting). Take care to avoid carryover contamination, and perform any analysis of amplified products away from the preparation area.

7. If necessary, optimize the reaction by testing concentrations of reaction components (e.g., RT enzyme) or the reaction temperature.

   *The amount of reverse transcriptase used can be very low (as little as 0.004 U/μl), but the RT enzyme typically does not contribute to non-template amplification. For more efficient*

*cDNA synthesis, lower reaction temperatures can be used, as commercially available RT enzymes are rapidly inactivated above 60°C.*

## MULTIPLEX LAMP REACTIONS USING DARQ PRIMERS

***ALTERNATE PROTOCOL 2***

LAMP reactions can be performed as above with multiple primer sets and targets, but require post-reaction processing for analysis (Aonuma et al., 2010; He and Lu, 2010; Liang et al., 2012). A recently developed method allows for real-time detection of multiple, distinct targets in single reactions using duplex LAMP primers, called DARQ (detection of amplification by release of quenching) probes (Tanner et al., 2012). As shown in Fig. 15.14.1B, the primers are identical in sequence to those used in standard LAMP reactions, except that the FIP or BIP is 5′-modified with a dark quencher (e.g., Iowa Black RQ), and is annealed to an additional 3′-modified oligonucleotide primer (Fd) that is complementary to F2 or B2c and contains a corresponding fluorophore (e.g., ROX). The primer duplex (qFIP:Fd) functions as a standard LAMP primer, and when the duplex region is encountered by the DNA polymerase, strand displacement removes Fd and releases the fluorescence signal by dequenching.

DARQ LAMP primers are designed using standard recommendations and software. DARQ requires no alterations to the primer design described for the standard LAMP process, other than modification of FIP (or BIP) and addition of the modified Fd sequence. The use of the reverse sequence (BIP) is recommended if the F2 sequence has a low $T_m$ (e.g., <60°C) or for troubleshooting problematic Fd:FIP pairs. The $T_m$ of the Fd oligonucleotide should be calculated using the reaction conditions and verified to be >60°C to ensure stable annealing during amplification. Two Fd examples are shown in the Materials list, and others can be found in Tanner et al. (2012). Modified oligonucleotides should be HPLC purified.

Be aware that if the fluorophore modification is directly adjacent to a guanine base, that base can be omitted from the primer sequence to avoid guanine quenching. However, this may slightly reduce the $T_m$ of the duplex. Therefore, if multiple guanines would need to be omitted from the sequence, consider switching the positions of quencher and fluorophore or using the reverse (B2) sequence.

The fluorophore selection must be matched to the available instrument used for detection. One fluorophore should be used per channel and assigned to one target primer set. Appropriate fluorophores and quenchers for the Bio-Rad CFX instrument listed below are presented in Table 15.14.1. Alternative fluorophores include the AlexaFluor and ATTO dyes, and alternative quenchers include BlackHole Quencher 1 and 2 and DABCYL. All oligonucleotides described herein were synthesized by Integrated DNA Technologies (IDT), and the available IDT modifications included numerous fluorophores and quenchers.

**Table 15.14.1** Examples of DARQ Primer Fluorophores and Quencher for the Bio-Rad CFX96 Instrument

| CFX Channel | | | Fluorophore | | | |
|---|---|---|---|---|---|---|
| | $\lambda_{ex}$ (nm) | $\lambda_{ex}$ (nm) | | $\lambda_{ex}$ (nm) | $\lambda_{em}$ (nm) | Quencher |
| 1 | 450-490 | 510-530 | 6-FAM | 495 | 520 | Iowa Black FQ (420-620 nm) |
| 2 | 515-535 | 560-580 | HEX | 538 | 555 | Iowa Black FQ |
| 3 | 560-590 | 610-650 | ROX | 588 | 608 | Iowa Black RQ (500-800 nm) |
| 4 | 620-650 | 675-690 | Cy5 | 648 | 668 | Iowa Black RQ |
| 5 | 672-684 | 705-730 | Cy5.5 | 685 | 706 | Iowa Black RQ |

***Additional materials*** *(also see Basic Protocol)*

Oligonucleotide primers (typically, 100 μM stocks in $H_2O$ or 1× TE buffer)

*Set 1 (e.g., for E. coli dnaE):*
F3: GATCACCGATTTCACCAACC
B3: CTTTTGAGATCAGCAACGTCAG
FIP: 5′-IAbRQ-
CTGCCCCGACGATAGGCTTAATCGTGGTCTGGTGAAGTTC
TACGG (Fd complement underlined)
Fd: ATTAAGCCTATCGTCGGGGCAG-Cy5-3′
BIP: TCCAGTGCGACCTGCTGGGTGGGTATTGTTCGCCGCCAGTAC
Loop F: TGCGCCATGTCCCGCT
Loop B: TGAGTTAACCCACCTGACG
*Set 2 (e.g., for HMBS2):*
F3: AGCAACATAGGTTTCAGTTC
B3: CTGTGAACGGTCATCACC
FIP: 5′-IAbRQ-
GATTCCACTTCCAACGTCGTTGCATAGGCATTGTATCCAGAGTG
(Fd complement underlined)
Fd: CAACGACGTTGGAAGTGGAATC-Rox_N-3′
BIP: CGAAGTGAACCTTGTCAACATGAGACTACCCACATCGTTACC
Loop F: ACGGACATGTCGATCATGGA
Loop B: CGTCTCCCTTCAATCCGATGGC

Control template DNA (e.g., *E. coli* genomic DNA, Affymetrix)
Intercalating dsDNA dye, e.g., SYTO-9 (see recipe; optional; for independent validation of amplification)
Multichannel fluorometer/thermocycler (e.g., Bio-Rad CFX-96)

***Pre-anneal DARQ duplexes***

1. Prepare a 1:1 mixture of labeled FIP and Fd. For 100 μM stocks, mix 50 μl FIP and 50 μl Fd to yield 100 μl of 50 μM duplex (sufficient for 250 reactions).

   *Opaque tubes can be used to minimize photobleaching, if desired.*

2. Heat mixture to 95°C, then allow to cool slowly to room temperature by removing from heat.

   *Annealed duplexes are stable for months at −20°C and can be repeatedly frozen/thawed.*

***Prepare primer mix and samples***

3. Prepare 100 μl of a 25× DARQ LAMP primer mix:

   20 μl 100 μM unmodified FIP
   40 μl 100 μM BIP
   5 μl 100 μM F3
   5 μl 100 μM B3
   10 μl 100 μM Loop F
   10 μl 100 μM Loop B
   10 μl $H_2O$ or 1× TE buffer.

   *The primer mix should be made using a 50% concentration of unmodified FIP to maintain reaction speed and overcome the slight inhibition caused by DARQ duplexes. The final reaction will contain 0.8 μM FIP and 0.8 μM qFIP:Fd (added separately). The DARQ duplex can be added to the other primers in one mixture, but to allow titration and optimization of the duplex concentration, the duplex and primer mixes are made separately here.*

4. Prepare samples as described (see Basic Protocol, steps 2 and 3).

***Perform LAMP reactions***

5. Prepare DARQ LAMP reactions on ice:

   For a 25-μl single-plex (one-target) amplification:

   12.5 μl 2× LAMP reaction buffer
   1 μl 25× primer mix
   0.4 μl 50 μM FIP:Fd duplex
   1 μl test sample or 10 ng control template DNA
   2 μl 8 U/μl *Bst* 2.0
   8.1 μl $H_2O$.

   For higher-plex reactions, keep the total primer concentration to 4.4 μM and scale each target set by 1/*n*, where *n* is the number of targets in the reaction (Tanner et al., 2012). For example, in a 25-μl duplex reaction combine:

   12.5 μl 2× LAMP reaction buffer
   0.5 μl 25× primer mix A
   0.2 μl duplex A
   0.5 μl 25× primer mix B
   0.2 μl duplex B
   1 μl sample A
   1 μl sample B
   2 μl 8 U/μl *Bst* 2.0
   7.1 μl $H_2O$.

   *Bst 2.0 is a more efficient LAMP enzyme when using DARQ primers (Tanner et al., 2012), and better performance can be seen by using higher amounts of DNA polymerase (unlike standard LAMP reactions). A good starting amount is 16 units in a 25-μl reaction.*

   *See Basic Protocol (step 4) for additional considerations, including replicates and controls.*

   *For an RNA target, reactions can be performed as above with addition of 0.5 μl AMV RT (see Alternate Protocol 1).*

   *If using DARQ to run internal control or standards, add serial dilutions of target across several test reactions.*

6. Mix reaction by pipetting or vortexing, then briefly pulse-spin in a microcentrifuge.

7. Place in a heated fluorescence instrument at 65°C. For a duplex with a $T_m$ significantly below this temperature (e.g., <60°C), the reaction may be performed at 63° or 60°C, although the efficiency may be reduced.

   *Fluorescence signal should appear with threshold times similar to or slightly slower than standard LAMP reactions. For multiplex reactions, threshold times will be slightly slower due to the reduced primer concentration, but will maintain the speed typical of LAMP reactions.*

   *To optimize reactions, the most efficient method is titration of the LAMP duplex. A 1:1 ratio of duplex to FIP (0.8 μM each for single-plex reactions) is a good initial condition. For increasing speed, duplex can be decreased to as little as 10% of the FIP reaction, although too little duplex will result in loss of signal.*

   *To independently measure LAMP amplification and DARQ signal generation, an intercalating dye can be included if its wavelength is distinct from that of the fluorophore on the duplex (e.g., SYTO-9 with Cy5 Fd). This allows one to determine whether overall LAMP efficiency is prohibitively reduced (in which case less duplex or more Bst2.0 should be used) or duplex is not being utilized (in which case more duplex should be used).*

8. Analyze data by determining threshold times for each target.

*Thresholds should be chosen independently for each fluorophore, as the signal will differ for each due to differential machine channels and quantum efficiencies.*

## REAGENTS AND SOLUTIONS

*Use molecular biology−grade water for all reagents and protocol steps, or RNase-free water when using RNA templates. For common stock solutions, see* **APPENDIX 2**; *for suppliers, see* **APPENDIX 4**.

### *Betaine solution, 5 M*

Can be purchased as a solution (Sigma) or made from solid form by dissolving in $H_2O$. If prepared, pass through a 0.22-μm filter. Store up to 2 years at 4°C.

### *Calcein solution, 625 μM*

For addition of calcein to LAMP reactions, calcein and manganese can be mixed into the 2× LAMP reaction buffer (see recipe) or added separately. To prepare the detection reagent mix, make a 625 μM stock of calcein (Sigma) in 12.5 mM $MnCl_2$ (25×). Store up to 1 year at −20°C.

### *dNTPs, 10 mM*

A mixture of dATP, dCTP, dGTP, and dTTP at 10 mM each can be purchased (e.g., New England Biolabs) or prepared from individual dNTPs (also from New England Biolabs). If made from dried solids, adjust prepare at pH 7.5 (at 25°C) and pass through a 0.22-μm filter. Stable for years at −20°C, or up to 1 year if repeatedly frozen/thawed.

### *Hydroxynaphtol blue, 3 mM*

Prepare a 3 mM stock of hydroxynaphthol blue (Sigma) in water (25×). Store up to 6 months at room temperature or 2 years at −20°C.

### *LAMP reaction buffer, 2×*

A 2× buffer can be prepared from individual components or by starting with Isothermal Amplification Buffer or Thermopol Buffer (New England Biolabs).

*From individual components*:
40 mM Tris, pH 8.8 (25°C)
20 mM $(NH_4)_2SO_4$
16 mM $MgSO_4$ (see recipe)
100 mM KCl
2.8 mM dNTPs (see recipe)
0.2% (v/v) Tween-20
Store in aliquots up to 6 months at −20°C
Vortex to resuspend after thawing

*From commercial buffer*:
200 μl 10× buffer
280 μl 10 mM dNTPs (see recipe)
120 μl 100 mM $MgSO_4$ (see recipe)
400 μl $H_2O$
Store in aliquots up to 6 months at −20°C
Vortex to resuspend after thawing

*Buffers and some components may precipitate during thawing, but will readily resuspend with vortexing. Buffers may be stored for many months and can undergo at least ten freeze/thaw cycles without loss of activity, but should nonetheless be dispensed in smaller aliquots for storage and use.*

*Isothermal Amplification Buffer (provided with Bst 2.0; New England Biolabs) contains 200 mM Tris, pH 8.8 (25°C), 500 mM KCl, 100 mM $(NH_4)_2SO_4$, 20 mM $MgSO_4$, and 1% (v/v) Tween 20. If prepared from components, pass through 0.22-μm filter. Store up to 3 years at −20°C, and vortex to resuspend after each thawing.*

*If starting from commercial buffers, the final (1×) KCl concentration will be 50 mM with Isothermal Amplification Buffer or 10 mM with Thermopol or Thermopol DF.*

*For real-time turbidity detection, Thermopol buffer should not be used for LAMP buffer due to the presence of Triton X-100, which interferes with turbidimetric detection. Instead, use Isothermal Amplification Buffer and Bst 2.0 or Thermopol Detergent Free (DF) and Bst LF.*

*For calcein detection, calcein and $MnCl_2$ can be included in the 2× buffer. If desired, add 50 μM calcein and 1 mM $MnCl_2$ to the above recipe.*

### *$MgSO_4$ solution, 100 mM*

If using commercial polymerase buffers (2 mM $Mg^{2+}$), additional $MgSO_4$ is necessary to achieve the 8 mM $Mg^{2+}$ final concentration in LAMP reaction buffer. A stock solution can be purchased (e.g., New England Biolabs) or made in-house. If made from dry powder, pass through 0.22-μm filter. Store up to 5 years at −20°C.

### *SYTO-9 solution, 50 μM*

Dilute 5 mM SYTO-9 stock (in DMSO; Life Technologies) 100× in water. Store in aliquots at −20°C in a closed box or opaque tubes. Stable for 6 months at −20°C.

*The diluted dye should be stored in aliquots, as loss of fluorescence can be seen with repeated freeze/thaw cycles.*

## COMMENTARY

### Background Information

The growing use of isothermal amplification methods has been aided by the ease of obtaining sequence information as well as increasing attention for point-of-care and field diagnostics. Several diagnostic isothermal methods employ more than one enzyme to create amplified products (reviewed in Niemz et al., 2011; Njiru, 2012). These employ a nicking enzyme (strand displacement amplification, SDA; nicking enzyme amplification reaction, NEAR) or recombination system for amplification (recombinase polymerase amplification, RPA), or a helicase to stimulate replication (helicase-dependent amplification, HAD; *UNIT 15.11*). One property is fundamental to these and other methods: strand-displacement activity of a thermophilic DNA polymerase, typically the large fragment of DNA polymerase I from *Bacillus stearothermophilus* (*Bst*). This property enables the use of a single reaction temperature, providing strand separation enzymatically rather than by heat denaturation as in PCR.

By design, LAMP requires only *Bst* or a similar DNA polymerase, an advantage for optimization, cost, materials procurement, and preparatory measures such as lyophilization (common for diagnostic applications). LAMP reactions employ four core primers recognizing six regions of target sequence, and are uniquely structured to produce a dumbbell-shaped loop structure for exponential amplification (see Fig. 15.14.1). Two auxiliary primers, the loop primers, can be added for increased amplification speed. The number of required target regions confers a high degree of specificity, and the strand-displacement activity of *Bst* in high magnesium buffer is sufficient to carry out rapid, robust amplification with DNA templates. Using new enzymes such as *Bst* 2.0, RNA targets can be amplified without need for a reverse transcriptase.

LAMP produces extensive amplification, such that detection of product may be performed using methods including agarose gel electrophoresis and intercalating dye fluorescence. This amount of amplification (>50× PCR yield) also enables detection by simple visual inspection using turbidity from precipitation of magnesium pyrophosphate (Mori et al., 2004), spectral shift from metal indicators (e.g., calcein, hydroxynaphthol blue; Tomita et al., 2008; Goto et al., 2009), probe-coupled nanoparticles (Seetang-Nun et al., 2013), or lateral flow devices (Murray et al., 2013). These simple methods enable LAMP reactions to be performed and detected using only a heat block or water bath. Although most LAMP tests remain in the laboratory, they are rapidly becoming an attractive diagnostic option. Isothermal methods have been described

for hundreds of pathogens and disease targets, including field targets (e.g., malaria, tuberculosis, HIV, trypanosomiasis) and clinical FDA-approved tests (e.g., *C. difficile* [LAMP], *C. trachomatis*, *N. gonorrhoeae*, and HSV [SDA]).

## Critical Parameters and Troubleshooting

### *Reaction conditions*

A key reaction component is the solution used to perform the amplification reaction, which contains two essential cofactors, $Mg^{2+}$ and dNTPs. The concentration of these components is important to obtain efficient amplification. The conditions listed in the protocol (8 mM $MgSO_4$, 1.4 mM dNTPs) are likely optimal for most reactions, but the $Mg^{2+}$ concentration can be adjusted as desired to change amplification efficiency and threshold (2 to 10 mM can be used successfully with varying results).

Addition of detergent improves performance of the DNA polymerase used, and detergent choice can be important, depending on detection methodology. For fluorescence and endpoint methods, either Triton X-100 or Tween 20 (provided in polymerase reaction buffer stocks) is suitable, but for real-time turbidity measurement Tween 20 should be used, as Triton X-100 can interfere with detection (Mori et al., 2004). If using the commercially provided polymerase buffers supplemented with $MgSO_4$ and dNTPs, Thermopol-DF (New England Biolabs) should be used in place of standard Thermopol in LAMP reactions using *Bst* large fragment. Isothermal Amplification Buffer, provided with *Bst* 2.0, uses Tween 20 and is suitable for all LAMP detection methods.

Temperature can also be a factor in optimizing LAMP reactions. Typically, best results for Bst, LF and Bst 2.0 will be seen at 65°C. However, in some cases Bst LF has been shown to perform more efficient LAMP at lower (60°–63°C) temperatures (Poole et al., 2012). To optimize reaction performance, it is advisable to perform a temperature gradient to identify peak amplification for the positive (desired, containing template) and lowest amplification for the negative (non-template, see below) controls. For Bst LF the optimal temperature will be 60°–65°C and Bst 2.0 63°–70°C.

### *LAMP primers*

The most important factors in any LAMP reaction are the core LAMP primers and the additional loop primers. Use of software for primer design (PrimerExplorer, LAMP Designer) is strongly recommended, as numerous factors contribute to the various actions of the primers, including overall $T_m$, local $T_m$ at the ends of the inner primers, AT content, and distance between the various target regions. Even with software-designed primers, the high number of variables leads to little predictive power about the primer set performance ab initio. For any new target, several primer sets (two to four) should be validated. Two key features are LAMP efficiency, measured by amplification speed and sensitivity, and specificity, measured by the degree of amplification in the absence of template nucleic acid. The primer set with best positive and lowest negative reaction should be chosen, and optimized if necessary by single base deletions at primer ends.

### *Non-template amplification*

A key criterion for diagnostic applications is specificity and minimization of false positives. Achieving this in LAMP is simple in some cases, but difficult with some primer sets. If new primers cannot be designed due to target restriction or locked-in sets, a few parameters can be adjusted to improve reaction performance. A common approach is to include additives, most prominently betaine. Concentrations of 0.4 to 1.2 M betaine have been used to improve specificity, although high amounts can impede positive reactions. Betaine is helpful if amplification is occurring in a set time window, in which case the positive reaction may slow upon addition of betaine, but the negative reaction (non-template amplification) can be pushed outside of the time window. Alternative parameters include temperature, which can be increased (to 70°C for *Bst* 2.0) to improve specificity; magnesium concentration, which is recommended at 8 mM, but can be tested at lower concentrations; enzyme concentration (as little as 0.02 U/µl *Bst* provides amplification); and time, with the rapidity of many positive reactions (under 10 to 15 min) allowing shortened incubation times and lower non-template false-positive amplifications. These factors are most effective with high copy number reactions, and if high sensitivity (<100 copy) detection is required and non-template amplification is high, primer redesign is likely the most effective strategy.

### *DARQ primer inhibition*

Use of the DARQ duplex LAMP primers introduces a varying amount of inhibition depending on sequence and stability of the duplex. Use of *Bst* 2.0 instead of wild-type

*Bst* large fragment minimizes this inhibition (Tanner et al., 2012). Another simple method for minimizing this inhibition is the use of unlabeled standard FIP along with the duplex, which provides for efficient LAMP reactions while maintaining high DARQ signal. With various numbers of multiplex and levels of inhibition, the ratio of duplex to FIP can be varied to optimize a specific reaction. Care should be taken to balance reaction speed with fluorescence signal, as the lowest amount of duplex likely provides the fastest reaction, but at the expense of fluorescence signal.

***LAMP product contamination***

Because amplification of LAMP structures is extremely efficient, reactions are sensitive to even minute amounts of product contamination (e.g., a $10^6$ dilution of a LAMP reaction can contain $10^4$ to $10^6$ molecules of LAMP amplicon). This property favors real-time monitoring or non-invasive examination of LAMP products (i.e., methods that do not involve opening of reaction vessels). For methods that require opening the tube or plate and extracting LAMP product, this should be performed in a space that is separate from where reactions are performed. If possible, a laminar flow hood and separate pipets and equipment should be used to avoid contamination. If contamination is suspected, all reagents should be replaced and areas and equipment treated with bleach.

***Sample considerations***

LAMP has been performed on a wide variety of sample types and has been demonstrated to have a higher tolerance to typical diagnostic specimen inhibitors than PCR (Kaneko et al., 2007). However, as with any nucleic acid amplification method, LAMP is sensitive to the presence of inhibitors in the sample, e.g., calcium, humic acid, and high amounts of whole blood, serum, or urine. LAMP is a good choice for diagnostic amplification in impure samples, but consideration should be given to the nature of the sample. Nucleic acid extraction is likely a component of most diagnostic workflows, and is recommended for LAMP reactions with clinical and agricultural samples.

## Anticipated Results

Expected results from any LAMP reaction are simple, namely a positive amplification in the presence of target nucleic acid. When using real-time detection methods, e.g., fluorescence or turbidimetry, a threshold time will be observed, with time increasing in inverse proportion to target concentration. Sensitivity depends strongly on the nature of the primer set used and its level of non-template amplification. An optimal primer set will provide clear distinction between low copy number (<10 copies) and non-template control, whereas a poor set may be indistinguishable at higher copy number (100 to 1000 copies) or fail to amplify in reasonable times with low template amounts (<1 fg small genome, 10 pg large). Typical thresholds can range from as fast as 3 to 5 min with efficient primers, optimal conditions, and high target amounts, to 45 to 60 min with suboptimal conditions, impure samples, or certain limited primer sets. Typically fluorescence will give 30% to 50% faster threshold times than turbidity due to the nature of the reaction, with turbidity thresholds ranging from 10 to 90 min. The highly exponential amplification reaction in LAMP results in a tightly ranged standard curve depending on efficiency of the reaction: an efficient LAMP reaction may vary by only 10 to 20 min over the $10^4$ to $10^8$ dynamic range, although this may represent 100% to 300% in relative terms. If using endpoint analysis, reactions should be incubated 15 to 90 min, depending on detection methods. Shorter times are sufficient for agarose gel electrophoresis, but longer times (typically, 45 to 60 min) should be used for methods like visual turbidity or color change (e.g., calcein, hydroxynaphthol blue). As described above, non-template amplification is of significant concern for any detection method, and care should be taken to choose an incubation time that provides detection of all desired target amounts but avoids non-template amplification.

## Time Considerations

LAMP reactions are typically very rapid and require very little time, with extraction of nucleic acid (if necessary) and analysis by gel electrophoresis representing the most significant time requirements. The standard LAMP method using DNA or RNA templates requires a small amount of preparatory time, including 10 to 30 min for primer design (necessary for each target, but only once) and 10 to 15 min for preparing LAMP buffer and reagents (stocks for >100 reactions can be prepared at once). At the time of the experiment, up to 2 hr are needed for sample preparation, depending on the nature of the sample, followed by 10 to 30 min setup time for one 96-well plate, depending on the complexity of the test and number of reaction variables. A typical LAMP reaction requires 10 to 90 min,

depending on the speed of amplification, target copy number, and level of non-template amplification. Many LAMP reactions can be completed in under 10 min. Reaction processing takes as little as 5 min for visualization of turbidity or fluorescence, but up to 1 to 2 hr for nanoparticle coupling or gel electrophoresis. In total, a typical LAMP reaction can be completed in ~90 min (30 min hands-on time). For DARQ LAMP, the only difference in time is the addition of a 10- to 20-min step for duplex annealing.

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